

4M20, Introduction to Robotics

Coursework - Part B

Soft robotics

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Q1 - Controlling Dielectric Elastomer Actuators

Dielectric elastomer actuators (DEAs) are a promising class of soft robotic actuators and artificial muscles capable of producing large strains from an input DC voltage.

Consider a DEA modelled as a cylinder of an incompressible and electrically insulating elastomer with thickness t . A voltage V is applied across its thickness using thin and stretchable plates covering its upper and lower surfaces, as shown in Figure 1. When $V = 0$, the dielectric thickness is t_0 . As the voltage is increased, the plates attract and the elastomer deforms. The variation in thickness t forced by the applied voltage V , minimises the (per unit volume) potential energy Π under static conditions.

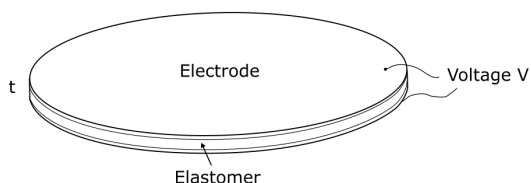

$$\Pi = \frac{\mu}{4} \left(\left(\frac{t}{t_0} \right)^4 + 2 \left(\frac{t_0}{t} \right)^2 - 3 \right) - \frac{\varepsilon_r \varepsilon_0}{2t^2} V^2 \quad [1]$$
$$t = t_0 (1 - KV^2)^{\frac{1}{6}} \quad [2]$$

Figure 1: A simplified DEA.

- (i) Given the expression for Π in Figure 1, which accounts for the energy of the DEA, derive the expression for t in Equation [2], explaining the key steps. Find an analytical expression for the constant K . Assume that ε_r , ε_0 , and μ are constant.
- (ii) The values in Table 1 are collected experimentally. Using these results and Equation [2], determine the value and units of K that best fits this data, and plot the relationship between t and V for $-30 \text{ kV} \leq V \leq 30 \text{ kV}$.

V (kV)	0	3.9	9.8	17.1	22.3	25.0	26.7	27.4	27.6
t (mm)	0.300	0.298	0.295	0.276	0.253	0.225	0.184	0.149	0.053

Table 1: Variation of DEA thickness with voltage.

- (iii) Consider the design of a DEA microactuator with a thickness stroke of 0.05 mm. Using the graph from part (ii), select ranges of applied voltage that are suitable for (a) precise and (b) rapid actuation. Motivate your selection.
- (iv) The elastomer is replaced with a material that has an identical graph, but a dielectric breakdown strength of 124 kV/mm. How would this knowledge change your answer to part (iii)?

Q2 - Error Sources in Stretchable Sensors

Uniaxial stretchable strain sensors change their resistance as they are extended. Due to their high compliance, they are often selected to monitor soft robotic proprioception, or for integration in wearable devices. 3 different resistive sensors are considered, labelled A, B, and C in Figure 2. The measured resistance $R_m = R + 2R_{cp}$ includes the sensor resistance R measured between the two contact points cp, and their contact resistances R_{cp} . $R = R_0$ corresponds to the unstretched length L_0 .

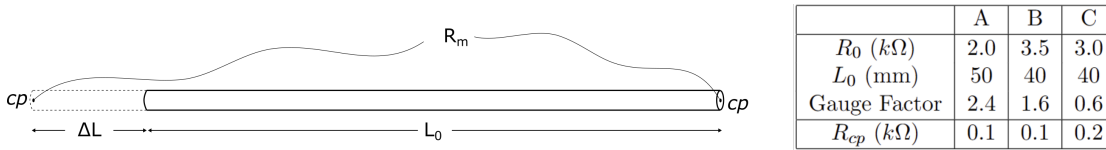


Figure 2: A stretchable resistive sensor.

- (i) Using the definition of Gauge Factor given in lectures, plot the measured resistance R_m of all 3 sensors for $0 \leq \Delta L \leq 20$ mm, first assuming that $R_{cp} \simeq 0$, then again using the tabled values in Figure 2.

During operation, the measured resistance R_m is used to predict ΔL . For each sensor, what is the prediction error if we assume $R_{cp} = 0$? Explain your answer.

- (ii) To measure R_m experimentally, a potential divider is used, such that the output voltage V_{out} depends on the input voltage V_{in} and a known resistance R_k . V_{out} is read using a data acquisition device which quantises its measurements M between 0 & 31:

$$V_{out} = \frac{R_k}{R_k + R_m} V_{in} \quad ; \quad M = \text{round} \left(31 \cdot \frac{V_{out}}{V_{in}} \right). \quad (1)$$

Using the tabled R_{cp} , $V_{in} = 5$ V, and $R_k = 10$ kΩ, explain which sensor would most accurately measure an extension of 10 mm.

Q3 - Fibre Optic Sensors

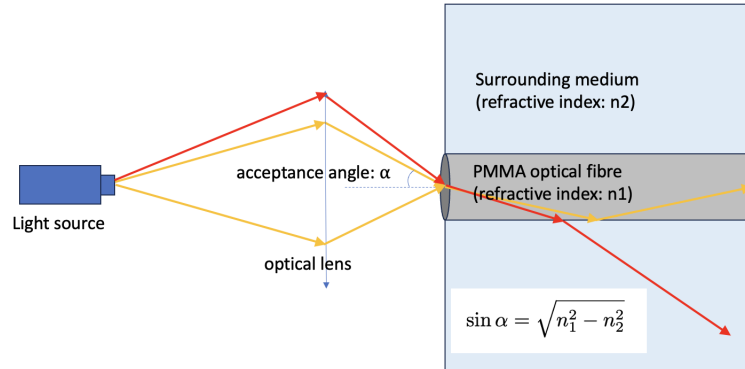


Figure 3: Acceptance angle (α) of a PMMA optical fibre in a medium. n_1 is the RI of the fibre, n_2 is the RI of the medium around the fibre.

Optical sensors can be used in soft robotics to monitor deformations through changes in light. The acceptance angle α is the maximal incident angle that the light can have to propagate inside the fibre, as shown in Figure 3.

- Consider the refractive index (RI) for the fibre at 1.49. Plot the acceptance angle given by external media whose RIs are in the range $[1, 1.5]$. Then, on the graph, mark the acceptance angles of the following external media: air (RI = 1), silicone rubber (RI = 1.42), and fused silica (RI = 1.46).
- Consider a light source focused onto the optical fibre with a maximum incident angle of 30° . Using Figure 1 in [2], identify which soft materials can be excluded by this sensor. Give and explain your answer.

Q4 Finite-element approximation of soft materials

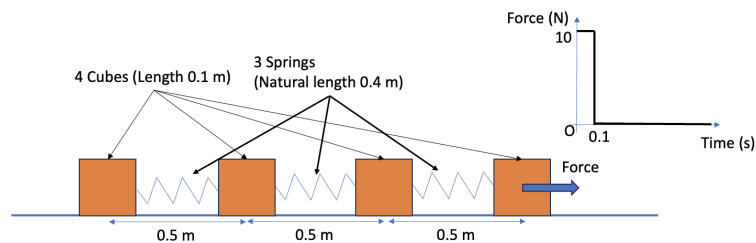


Figure 4: Finite-element model.

Use Simscape Multibody to simulate a finite-element approximation of a soft material. Starting from the Simscape file provided, modify it to build and simulate the following model.

- (i) Connect 4 identical cubes (length 0.1m, mass 1kg) with 3 springs (stiffness 10N/m, natural length 0.4m). The spring terminals should be connected to the external surface of each cube, as shown in Figure 4. The initial distance between the centres of the cubes should be 0.5m.
- (ii) Apply a pulse-like force (magnitude 10N, duration 0.1s) to the right cube, as shown in Figure 4. Plot the trajectory of each cube over time, within an interval of 10 seconds.

Explore how different values for the spring stiffness affect the behaviour of the cubes. Discuss your findings in the report, supporting your argument with additional plots.

- (iii) The finite-element model is highly oscillatory. Reduce oscillations by adding dampers between each pair of cubes. Explore the effect of several damping coefficients on the overall behaviour of the cubes. Discuss your findings in the report, supporting your argument with additional plots.

Bibliography

- [1] Pelrine et al., High-Speed Electrically Actuated Elastomers with Strain Greater Than 100%. Vol 287, Issue 5454, pp. 836-839 (2000).
- [2] Kim et al., Refractive index of soft contact lens materials measured in packaging solution and standard phosphate buffered saline and the effect on back vertex power calculation. Contact Lens and Anterior Eye Vol. 43, Issue 2, pp. 123-129 (2020)

Your Report

- Please refer to the Moodle website for the exact submission deadline for this report. Be aware that penalties will be applied to all late submissions

Undergraduates must use only their Coursework Candidate Number (CCN) and must not include their name in the report or file name. MPhil and postgraduate students must include their name and CRSID on their reports.

- Please submit your report in the preferred four-page, double-column format. Templates for both LaTeX and Microsoft Word are available on the Moodle website. Your report should be a concise exposition that focuses on the salient features of your findings. Ensure that each figure is properly annotated and directly contributes to your explanations. While the evaluation will be based solely on the four pages of the report, you are welcome to include an appendix for any additional materials.
- Please upload an archive file with your code and/or simulation files.
- This assignment is worth 25% of your final grade, with an expected time commitment of approximately 10 hours. For any questions regarding this assignment, please contact Xiang Wang (xw438), David Hardman (dsh46), and Fulvio Forni (ff286).